

Robust beamforming anti-mainlobe jamming method based on MIMO-STCA radar

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Abstract—To address the issue of degraded jamming suppression performance in space-time coding array (STCA) beamforming under range quantization error conditions, we propose a robust beamforming method for mainlobe jamming suppression based on STCA radar. Firstly, the transmit-receive signal model of the STCA is presented. By Combining beampattern control theory, we introduce a beampattern control algorithm based on the minimum peak sidelobe criterion (BC-MPS), achieving joint optimization of multiple null regions and the mainlobe region. Furthermore, we design a new robust beamforming method for mainlobe jamming suppression based on the proposed beampattern control algorithm. Compared to existing algorithms, the proposed method can achieve a more stable mainlobe and deeper null regions, effectively suppressing multiple mainlobe range false target jammings under range quantization error conditions. Finally, simulation experiments validate the effectiveness and accuracy of the proposed method.

Keywords—Space-time coding array, MIMO-STCA radar, robust beamforming, anti-mainlobe jamming.

I. INTRODUCTION

Due to the two-dimensional dependency of transmit beampattern on range and angle, space-time coding arrays exhibit excellent performance in resisting mainlobe jamming. However, in practical scenarios, the impact of range quantization errors degrades the mainlobe jamming suppression performance. This paper combines beampattern control theory and proposes a robust beamforming method for space-time coding radar arrays [1],[2],[3],[4],[5],[6] to mitigate the performance degradation caused by range quantization errors.

To achieve array beampattern control, numerous researchers have conducted extensive research. Numerical synthesis algorithms construct the cost function of the beampattern control problem based on the error between the designed beampattern and the target beampattern [7],[8]. Beampattern control is achieved by iteratively updating the covariance matrix. However, this method uses a trial-and-error approach to calculate the virtual jamming power, failing to achieve precise array response control. Global optimization algorithms solve the beampattern control problem using optimization algorithms such as ant colony optimization, genetic algorithms, and simulated annealing [9],[10]. Nonetheless, these algorithms do not guarantee optimal beampattern performance. Convex optimization algorithms convert the beampattern control problem into a convex expression and solve it using interior-point methods [11].

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Issues like flat-top beam synthesis and robust sidelobe suppression, which are non-convex in nature, can be solved using semidefinite programming and semidefinite relaxation methods [12],[13],[14]. The core idea of iterative Fourier algorithms is to utilize the Fourier transform relationship between the weighting vector and the far-field beampattern, obtaining the array response through fast Fourier transform and inverse fast Fourier transform [15],[16]. Array response control algorithms achieve precise control of the array response in the azimuth to be controlled by superimposing a designed auxiliary beampattern on the initial beampattern [17],[18],[19],[20],[21].

Based on precise array beampattern control theory, this paper proposes a beampattern control algorithm with the minimum peak sidelobe criterion (BC-MPS). Firstly, by adopting the concept of adaptive beamforming and conducting optimal weight vector analysis, it is discovered that accurate control of the beampattern magnitude in a specific direction can be achieved by reasonably designing the parameter ζ . Further, by decomposing the weight vector into an iterative form, accurate control of the beampattern magnitude in a specific region can be realized through iterative solving of multiple parameters ζ_k . Subsequently, to determine the specific value of parameter ζ , a simple and intuitive geometric method is selected to solve for the values of parameter ζ , obtaining a set of solutions. Finally, based on the minimum peak sidelobe criterion, the optimal solution for parameter ζ is calculated. The proposed beampattern control algorithm is then applied to space-time coding array radar, forming a beampattern with a flat-top mainlobe and wide nulls in the transmit and receive two-dimensional spatial frequency domain. It improves the robustness of the parameter design for the beamforming method against mainlobe jamming.

II. SIGNAL MODEL OF MIMO-STCA RADAR

The array layout of the MIMO-STCA (Multiple-Input Multiple-Output Space-Time Coding Array) radar system is shown in a co-located planar array that adopts a rectangular grid structure with M rows and N columns, and the inter-element spacing is half-wavelength. The time shift Δt is introduced between rows of transmit signals, and the time delay of the m -th row transmit array is

$$\tau_m^T = (m-1)\Delta t, \quad m=1,2,\dots,M, \quad (1)$$

The transmit signal of each element in the m -th row can be written as $s_m(t) = e^{j2\pi f_0 t} \varpi_m(t - \tau_m^T)$, where f_0 is the carrier

frequency, $\varpi_m(t) = c_m g(t)$ is the orthogonal signal and satisfies the following equation

$$\int_{T_p} \varpi_m(t) \varpi_\ell(t) dt = 0, \forall m, \ell = 1, 2, \dots, M, m \neq \ell \quad (2)$$

where c_m is the orthogonal coding, and

$$g(t) = \text{rect}\left(\frac{t}{T_p}\right) \exp(j\pi\mu t^2) \quad (3)$$

where $\text{rect}(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & \text{else} \end{cases}$, T_p is the radar pulse duration, $\mu = B/T_p$ is the chirp rate, B is the signal bandwidth.

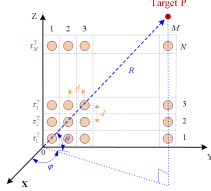


Fig. 1. MIMO-STCA radar

We assume a point target P in far-field at elevation angle θ , azimuth angle φ and range R . By taking the array element at the coordinate origin as the reference element, the unit vector in the direction of target P can be written as (r_x, r_y, r_z) , where $r_x = \cos\theta \cos\varphi$, $r_y = \cos\theta \sin\varphi$ and $r_z = \sin\theta$.

The spatial phase shift between the transmit signal of element m -th row, n -th column ($n=1, 2, \dots, N$) and the reference element is caused by two factors: the difference in propagation range and the directional alignment toward target. This relationship is formulated as $\alpha_{(m,n)} = (n-1)\alpha_y + (m-1)\alpha_z$, where $\alpha_y = 2\pi d \cos\theta \sin\varphi / \lambda$ and $\alpha_z = 2\pi d \sin\theta / \lambda$ represent the phase shift caused by the path difference between adjacent elements in the horizontal and vertical directions, respectively, and λ is the wavelength.

The superimposed signal $S(t)$ of the transmit signals from all elements in space can be expressed as

$$S(t) = \sum_{n=1}^N \sum_{m=1}^M e^{j(n-1)\alpha_y} e^{j(m-1)\alpha_z} S_m(t) \quad (4)$$

$$= e^{j2\pi f_d t} \sum_{n=1}^N \sum_{m=1}^M e^{j(n-1)\alpha_y} e^{j(m-1)\alpha_z} c_m e^{j\pi\mu(t-\tau_m)^2}$$

Accordingly, under the assumption that the transmit signal is a narrowband signal, the target echo signal received by the m' -th row and n' -th column array element, after the radar transmit signal is reflected by target P, can be represented as

$$y_{(m',n')}(t) = e^{j(n'-1)\alpha_y} e^{j(m'-1)\alpha_z} S(t-\tau) \quad (5)$$

$$= e^{j2\pi f_d(t-\tau)} e^{j(n'-1)\alpha_y} e^{j(m'-1)\alpha_z} \sum_{n=1}^N \sum_{m=1}^M e^{j(n-1)\alpha_y} e^{j(m-1)\alpha_z} c_m e^{j\pi\mu(t-\tau-\tau_m)^2}$$

where $\tau = 2R/c$ is the common time delay, and c is the speed of light.

Next, after down-conversion processing and digital mixing correlated with Δt , the processed echo in the m' -th row and n' -th column element can be written as

$$\tilde{y}_{(m',n')}(t) = \hat{y}_{(m',n')}(t) e^{-j2\pi f_d \Delta t} e^{j2\pi\mu\Delta t\tau_m} e^{-j2\pi\mu\tau_m^2} \quad (6)$$

$$= e^{-j2\pi f_d \Delta t} e^{j(n'-1)\alpha_y} e^{j(m'-1)\alpha_z} \sum_{n=1}^N \sum_{m=1}^M e^{j(n-1)\alpha_y} e^{j(m-1)\alpha_z} c_m \varpi_m(t-\tau) e^{j2\pi\mu\tau_m}$$

Then, the matched filter designed for the transmit signal of the m -th row is expressed as $h_m(t) = \varpi_m^*(-t)$ and the echo for each receive channel passes through a bank of matched filters on each receive channel, where the output of the m' -th row and n' -th column receive signal from the m -th filter, is written as

$$y_{(m',n',m)}(t) = \tilde{y}_{(m',n')}(t) * h_m(t) \quad (7)$$

$$= \beta e^{j(n'-1)\alpha_y} e^{j(m'-1)\alpha_z} e^{j2\pi\mu\tau_m} + e^{-j2\pi f_d \Delta t} e^{j(n'-1)\alpha_y} e^{j(m'-1)\alpha_z} \sum_{n=1}^N \sum_{m=1}^M e^{j(n-1)\alpha_y} e^{j(m-1)\alpha_z} e^{j2\pi\mu\tau_m} \int_{T_p} \varpi_l(x-\tau) \varpi_m^*(x-t) dx$$

where $\beta = e^{-j2\pi f_d \Delta t} \sum_{n=1}^N e^{j(n-1)\alpha_y}$, since $\varpi(t)$ is an orthogonal signal, the equation (7) can be written as

$$y_{(m',n',m)}(t) = \beta e^{j(n'-1)\alpha_y} e^{j(m'-1)\alpha_z} e^{j(m-1)\alpha_z} e^{j2\pi\mu\tau_m} e^{j2\pi\mu\tau_m} \quad (8)$$

namely the echo signal expression of all transmit signals in the m -th row received by the receive element in the m' -th row and n' -th column.

Next, the output signal $y_{(m',n')}$ of the receive echo signal in m -th row and n -th column after M-dimensional matched filtering is written as

$$y_{(m',n')}(t, \varphi, R) = [y_{(m',n',1)}(t, \varphi, R), \dots, y_{(m',n',M)}(t, \varphi, R)] \quad (9)$$

Finally, all signals after matched filtering processing can be written as y , which is a column vector of dimension $M^2 N \times 1$, denoted as

$$y(t, \varphi, R) = \begin{bmatrix} y_{(1,1)}, y_{(1,2)}, \dots, y_{(1,N)} \\ y_{(2,1)}, y_{(2,2)}, \dots, y_{(2,N)} \\ \vdots \\ y_{(M,1)}, y_{(M,2)}, \dots, y_{(M,N)} \end{bmatrix} \quad (10)$$

$$= \beta \mathbf{a}_y(\theta, \varphi) \otimes \mathbf{a}_z^T(\theta, R) \otimes \mathbf{a}_z^R(\theta)$$

where, $\otimes$ represent Kronecker product, $\mathbf{a}_y(\theta, \varphi)$ is horizontal steering vector, $\mathbf{a}_z^T(\theta, R)$ is vertical transmit steering vector, $\mathbf{a}_z^R(\theta)$ is vertical receive steering vector, and their expressions are as follows

$$\mathbf{a}_y(\theta, \varphi) = [1, e^{j(N-1)2\pi d \cos\theta \sin\varphi / \lambda}, \dots, e^{j(N-1)2\pi d \cos\theta \sin\varphi / \lambda}]^T \quad (11)$$

$$\mathbf{a}_z^T(\theta, R) = [1, e^{j2\pi d \sin\theta / \lambda} e^{j4\pi\mu\Delta t\tau_1 R/c}, \dots, e^{j(M-1)2\pi d \sin\theta / \lambda} e^{j(M-1)4\pi\mu\Delta t\tau_1 R/c}] \quad (12)$$

$$\mathbf{a}_z^R(\theta) = [1, e^{j2\pi d \sin\theta / \lambda}, \dots, e^{j(M-1)2\pi d \sin\theta / \lambda}]^T \quad (13)$$

It can be seen from the above formula that for MIMO-STCA radar, its horizontal steering vector is only a function of the angle, while the vertical steering vector is a bivariate function of both angle and range.

III. ROBUST MAINLOBE JAMMING SUPPRESSION WITH MIMO-STCA RADAR

Consider mainlobe range false target jamming generated by a throwing decoy that intercepts and repeats the radar signal. The decoy generates Q false target whose angles are identical to the true target but with different ranges. Assume that the location of the true target is $(\theta_s, \varphi_s, R_s)$ and the locations of the

q -th ($q=1,2,\dots,Q$) false target jamming is $(\theta_q, \varphi_q, R_q)$, then $\theta_q = \theta_s$, $\varphi_q = \varphi_s$, $R_q \neq R_s$, and the echo signal of the q -th false target can be similarly expressed as

$$\mathbf{y}_q(\theta_q, \varphi_q, R_q) = \beta_q \mathbf{a}_y(\theta_q, \varphi_q) \otimes \mathbf{a}_z^T(\theta_q, R_q) \otimes \mathbf{a}_z^R(\theta_q, R_q) \quad (14)$$

where $\mathbf{y}_q \in \mathbb{C}^{M^2 N}$.

Accordingly, all the receive signal vector of the MIMO-STCA can be written as $\mathbf{z} = \mathbf{y}(\theta_s, \varphi_s, R_s) + \sum_{q=1}^Q \mathbf{y}_q(\theta_q, \varphi_q, R_q) + \mathbf{n}$, where $\mathbf{n} \in \mathbb{C}^{M^2 N \times 1}$ is modeled as zero-mean white Gaussian noise.

To achieve mainlobe jamming suppression, this paper draws inspiration from adaptive beamforming and proposes a novel method for designing the vertical weight vector $\mathbf{w} \in \mathbb{C}^{M^2}$ for robust beamforming. The weight vector $\mathbf{w}$ is written as $\mathbf{w}_z = \mathbf{w}_{zR} \otimes \mathbf{w}_{zT} \in \mathbb{C}^{M^2}$, where $\mathbf{w}_{zR} = \mathbf{a}_{zR}(\theta_s) \in \mathbb{C}^M$. According to the minimum variance distortionless response criterion, an optimal weight vector can be constructed as the following expression

$$\tilde{\mathbf{w}}_{zT} = \beta \mathbf{R}_{i+n}^{-1} \mathbf{a}_{zT}(\theta_s, R_s) \in \mathbb{C}^M \quad (15)$$

where $\mathbf{R}_{i+n}$ is the jamming-plus-noise covariance matrix.

Considering a single jamming scenario, the covariance matrix of the jamming plus noise can be expressed as

$$\mathbf{R}_{i+n} = \alpha_i^2 \mathbf{a}_z^T(\theta_i, R_i) \mathbf{a}_z^T(\theta_i, R_i) + \alpha_n^2 \mathbf{I}_M \quad (16)$$

where, α_i^2 is power of jamming, α_n^2 is power of noise, $\mathbf{a}_z^T(\theta_i, R_i)$ is vertical steering of jamming, $\mathbf{I}_M \in \mathbb{C}^M$ is identity matrix.

By applying the matrix inversion rule and substituting equation (16) into equation (15), the optimal weight vector is expressed as

$$\tilde{\mathbf{w}}_{zT} = \frac{\mu}{\alpha_n^2} \left(\mathbf{a}_{zT}(\theta_s, R_s) - \frac{\frac{\alpha_i^2}{\alpha_n^2} \mathbf{a}_{zT}(\theta_i, R_i) \mathbf{a}_{zT}^H(\theta_i, R_i) \mathbf{a}_{zT}(\theta_s, R_s)}{1 + \frac{\alpha_i^2}{\alpha_n^2} \|\mathbf{a}_{zT}(\theta_i, R_i)\|_2^2} \right) \quad (17)$$

As μ/α_n^2 does not affect the performance of the vertical transmit beampattern, this term is ignored, and $\tilde{\mathbf{w}}_{zT}$ is written as $\tilde{\mathbf{w}}_{zT} = \mathbf{w}_0 + \zeta \mathbf{a}_z^T(\theta_i, R_i)$, where $\mathbf{w}_0 = \mathbf{a}_z^T(\theta_s, R_s)$ is the initial weight vector.

and the expression for ζ is

$$\zeta = \frac{\frac{\alpha_i^2}{\alpha_n^2} \mathbf{a}_z^H(\theta_i, R_i) \mathbf{a}_z^T(\theta_s, R_s)}{1 + \frac{\alpha_i^2}{\alpha_n^2} \|\mathbf{a}_z^T(\theta_i, R_i)\|_2^2} \quad (18)$$

According to reference [16], it is known that by varying the parameter ζ , the normalized power of the vertical transmit beampattern at (θ, R_i) can be controlled. To control the power of the beam pattern at multiple points, the vertical transmit steering vector is designed as $\mathbf{w}_k = \mathbf{w}_{k-1} + \zeta_k \mathbf{a}_{zT}(\theta_k, R_k)$, where k is the iteration count. Furthermore, the normalized magnitude response of the vertical transmit beampattern is written as

$$A_k(\theta_k, \theta_s, R_k, R_s) = \frac{|\mathbf{w}_k^H \mathbf{a}_{zT}(\theta_k, R_k)|^2}{|\mathbf{w}_k^H \mathbf{a}_{zT}(\theta_s, R_s)|^2} = \rho_k \quad (19)$$

where $\rho_k = \eta_k^2 e^{j\phi_k}$ represents the normalized power response at θ_k , η_k and ϕ_k represents the normalized amplitude and the normalized phase at θ_k .

Furthermore, the expression for the parameter ζ_k is obtained using geometric methods

$$\zeta_k(\phi_k) = -\frac{\mathbf{Q}_k^*(1, 2)}{\mathbf{Q}_k(2, 2)} + \frac{\sqrt{-\det(\mathbf{Q}_k)}}{|\mathbf{Q}_k(2, 2)|} e^{j\phi_k} \quad (20)$$

$$\mathbf{Q}_k = \begin{bmatrix} \mathbf{w}_{k-1} \\ \mathbf{a}_{zT}(\theta_k, R_k) \end{bmatrix}^H \begin{bmatrix} \mathbf{w}_{k-1} & \mathbf{a}_{zT}(\theta_k, R_k) \end{bmatrix} \\ \bullet (\mathbf{a}_{zT}(\theta_k, R_k) \mathbf{a}_{zT}^H(\theta_k, R_k) \\ - \rho_k \mathbf{a}_{zT}(\theta_s, R_s) \mathbf{a}_{zT}^H(\theta_s, R_s)) \quad (21)$$

Thus the solution for parameter ζ_k can be transformed into the solution for parameter ϕ_k .

Hence, in order to obtain an appropriate ϕ_k , based on the minimum peak sidelobe criterion (MPS), the following beampattern control (BC) optimization problem is formulated as

$$\begin{cases} \min \max_{\phi_k \in [0, 2\pi] | (\theta, R) \in \Theta_m} |\mathbf{w}_k^H \mathbf{a}_{zT}(\theta, R)| \\ s.t. \begin{cases} \mathbf{w}_k = \mathbf{w}_{k-1} + \zeta_k(\phi_k) \mathbf{a}_{zT}(\theta_k, R_k) \\ \mathbf{w}_k \mathbf{a}_{zT}(\theta_s, R_s) = 1 \end{cases} \end{cases} \quad (22)$$

where θ_s is the direction pointed to by the mainlobe, and Θ_m is the region of the mainlobe.

By solving the optimization problem given in equation (22), one can obtain the optimal value ϕ_k^{opt} for ϕ_k , and further derive the optimal vertical transmit weight vector

$$\mathbf{w}_k = \mathbf{w}_{k-1} + \left(-\frac{\mathbf{Q}_k^*(1, 2)}{\mathbf{Q}_k(2, 2)} + \frac{\sqrt{-\det(\mathbf{Q}_k)}}{|\mathbf{Q}_k(2, 2)|} e^{j\phi_k^{opt}} \right) \mathbf{a}_{zT}(\theta_k, R_k) \quad (23)$$

When performing beamforming, the angular region is divided into the mainlobe region Θ_m and the sidelobe region Θ_s . To achieve robust beamforming, the vertical transmit array pattern needs to satisfy the following three conditions: (1) The mainlobe of the beampattern points to the azimuth angle where the target is located. (2) The error of the mainlobe between the beampattern and desired beampattern is less than the preset error. (3) The sidelobe region of the beampattern is less than the preset threshold. Notice that the sidelobe region includes both the null region Θ_s^0 and the non-null region Θ_s^1 .

According to conditions (1)-(3), in the k -th iteration, the azimuth angles to be controlled for the mainlobe region and the sidelobe region are respectively determined as

$$(\theta_{m,k}, R_{m,k}) = \arg \max_{(\theta, R) \in \Theta_m} |A_{w_{k-1}}(\theta, \theta_s, R, R_s) - \rho_m| \quad (24)$$

$$(\theta_{s,k}, R_{s,k}) = \arg \max_{(\theta, R) \in \Theta_s^0} |A_{w_{k-1}}(\theta, \theta_s, R, R_s) - \rho_n| \quad (25)$$

where ρ_m and ρ_n are respectively denoted as the normalized power response for the desired beampattern of the mainlobe and the sidelobe.

Given a maximum deviation threshold ϑ_m for the mainlobe region and a preset threshold ϑ_n for the null region, if the beampattern meets the design requirements, then the k -th iteration should satisfy the following conditions

$$\max |A_{w_k}(\theta, R, \theta_0, R_0) - \rho_m| \leq \vartheta_m, (\theta, R) \in \Theta_m \quad (26)$$

$$\max (A_{w_k}(\theta, R, \theta_0, R_0) - \rho_n) \leq \vartheta_n, (\theta, R) \in \Theta_s^0 \quad (27)$$

The termination criteria in (27) are governed by preset error tolerances. Specifically, the mainlobe deviation threshold is determined by the maximum allowable gain fluctuation from range quantization errors, while the null region threshold is based on the required jamming suppression ratio. The process terminates when both precision requirements are met, ensuring simultaneous mainbeam robustness and effective interference rejection.

According to equation (12) and (13), the vertical transmit spatial frequency is expressed as

$$f_{zT}(\theta, R) = 2\mu R \Delta R_{Rx} / c + d \sin \theta / \lambda \quad (28)$$

$$f_{zR}(\theta, R) = d \sin \theta / \lambda \quad (29)$$

Therefore, when (θ_k, R_k) is determined, the corresponding transmit spatial frequency is written as $f_{zT}(\theta_k, R_k)$.

IV. SIMULATION RESULTS

Furthermore, it is worth noting that the proposed BC-MPS algorithm exhibits inherent robustness against typical array manifold perturbations, such as amplitude and phase errors. By proactively broadening the mainlobe and widening the null regions via constraints $\mathcal{G}_m$ and $\mathcal{G}_n$, this "design redundancy" provides a physical safety margin. Even in the presence of minor amplitude/phase deviations, the interference remains trapped within the widened null region, preventing the "mainlobe self-cancellation" risk commonly faced by traditional adaptive methods.

To validate the effectiveness of the proposed robust anti-mainlobe jamming method, we conduct numerical simulations with the following parameter configuration. The normalized vertical transmit and receive spatial frequencies are both defined within the range of $[-0.5, 0.5]$. The true target is positioned at broadside with azimuth angle $\phi_0=0^\circ$ and elevation angle $\theta_0=0^\circ$. After range compensation processing, its corresponding vertical transmit and receive spatial frequencies become $f_{zT}^s=0$ and $f_{zR}^s=0$ respectively. For false targets 1 and 2, the vertical transmit spatial frequencies are set to $f_{zT}^1=0.3$ and $f_{zT}^2=-0.4$ respectively, while the vertical receive spatial frequencies are set to $f_{zR}^1=0$ and $f_{zR}^2=0$. Fig.2 shows the distribution of true and false targets in the 2D spatial frequency domain. Due to range quantization errors, the actual target positions deviate from theoretical ones, necessitating robust anti-mainlobe jamming methods to mitigate error effects.

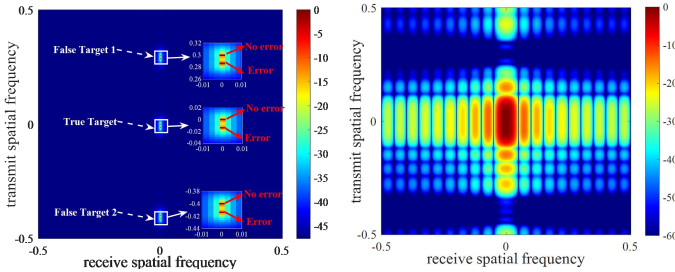


Fig. 2. True and false target distribution.

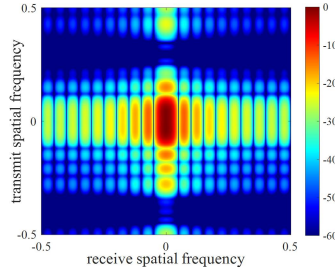


Fig. 3. Robust Beampattern method based on BC-MPS.

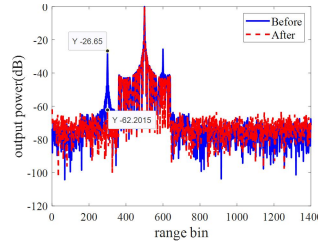


Fig. 4. Output power with robust anti-mainlobe jamming method.

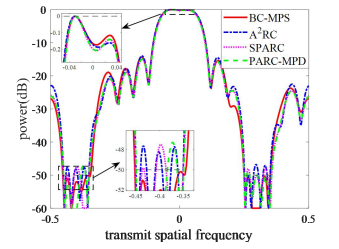


Fig. 5. Beampattern performance for different algorithms.

Fig. 3 presents the robust beamforming pattern in the vertical dimension obtained using the proposed algorithm. It is evident that a flat mainlobe is formed at the position of the true target, while a wide null is formed at the position of the false target. This effectively overcomes the problem of decreased jamming suppression performance caused by errors. Fig. 4 compares the pulse compression output power of conventional and robust beamforming under error conditions, demonstrating that the robust method effectively suppresses jamming and improves the suppression ratio by 30 dB. Fig. 5 compares BC-MPS with other robust algorithms (A2RC, SPARC, and PARC-MPD), showing it forms a smoother mainlobe and deeper nulls. The precision of interference suppression is dictated by the spatial degrees of freedom, which depend on the vertical array element count N . Simulations confirm that an adequate array scale allows the efficient, simultaneous realization of a flat-top mainlobe and deep nulls. Furthermore, compared to classical sample matrix inversion (SMI) methods — which suffer from sample contamination and mainlobe self-cancellation under mainlobe jamming — BC-MPS maintains a stable mainlobe gain and superior jamming suppression.

For real-time feasibility, BC-MPS drastically reduces its iterative computational load. Unlike A2RC's non-convex optimization or SPARC's high-dimensional operations, BC-MPS uses a simple geometric method to determine μ , avoiding matrix inversions and SDR processes. It converges within dozens of iterations, yielding significantly shorter processing times while maintaining a 30 dB jamming suppression improvement.

V. CONCLUSIONS

To mitigate the performance degradation of conventional beamforming-based mainlobe jamming suppression methods, this study proposes a robust beamforming framework utilizing the BC-MPS algorithm in STCA radar systems. Experimental validation demonstrates that the proposed method achieves superior jamming suppression performance with 30 dB enhancement in jamming suppression ratio, while maintaining stable performance under severe range quantization error conditions. This advancement significantly improves the robustness of mainlobe jamming suppression methodologies in STCA radar architectures, particularly in scenarios involving high-resolution range quantization and complex electromagnetic environments.

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